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Chapter 1: What are Computers?
Q1. 10 daily life examples of 3D printing:
1. Clear dental aligners (transparent ~~toth~~ tooth straightening molds).
2. Waterproof breathable arm casts (plastic mesh for fractures).
3. Custom spectacle frames (tailored to facial dimensions).
4. Running shoe midsoles (Adidas 4DFWD honeycomb lattice).
5. Custom hearing aid shells (shaped to ear canal scans).
6. Replacement home ~~aptiance~~ appliance knobs and fridge shelf clips.
7. 1:1 scale anatomical organ models for surgical rehearsal.
8. Rapid concrete houses (walls extruded in 48–72 hours).
9. Lost-wax jewelry casting molds with micro details.
10. Ergonomic ~~uterosli~~ utensil grips for patients with arthritis.

Q2. Programmable vs Non-Programmable machines:
• Non-Programmable: Hardwired physical function (fan, toaster, clock). Cannot alter behavior

without rebuilding hardware parts.
• Programmable: CPU + RAM separate from software instructions. Fetches changing instructions from memory to switch tasks instantly.

Q3. Childhood dream (Astronaut / Space Exploration):
(a.) Today: Spacecraft on autopilot. SpaceX Dragon docks autonomously; Mars rovers drive around obstacles using computer vision.
(b.) B.Tech S+AI: Gives skills in vision, robotics, and telemetry to write autonomous navigation software for space probes.
(c.) Closer? Yes. Space is largely a software challenge today. Writing spacecraft code makes me a direct part of the mission.

Chapter 2: Hardware + Software
Q1. Chip designing video summary (Branch Education):
1. Silicon Wafer: Quartz sand refined to 99.9999999% silicon boule, sliced into razor-thin mirror wafers.
2. Photolithography: EUV light (13.5 nm) projected through a photomask stencil to etch nanometer-scale circuit layouts.
3. Doping: Boron/Phosphorus ion implantation creates NP channels to form MOSFET switching transistors.
4. Interconnects: 30-50 copper wiring layers deposited on top.

5. Packaging: Wafer diced into dies, tested, and heat-spread sealed.

Q2. Why did RAM get expensive in recent years?

- AI Boom + HBM: High demand for AI GPUs (H100) diverted cleanrooms to HBM3e, starving DDR4/DDR5 consumer PC supply.
- 3-Firm Oligopoly: Samsung, SK Hynix, Micron (90%+ market) cut wafer production to enforce price discipline.
- Tooling ~~Capex~~ Capex: Sub-10nm DRAM requires $2-00M+ ASML EUV machines.

Chapter 3: CLI

1. Show all lines of a file except last 5:
Ans: head -n -5 filenametxt (negative skips trailing lines)
2. Find files bigger than x MB in home:
Ans: find -type f -size +50M
(= home, +50M = >50MB)
3. 10 largest files in /var/log:
Ans: find /var/log -type f -exec du -h {} +
2>/dev/null | sort -hr | head -n 10
4. Count jpg files under current folder:
Ans: ~~fcrcd~~ find . -type f -iname "*.jpg" | wc -l
(find piped to wc -l)
5. Count words in a file:
Ans: wc -w notestxt (-w isolates whitespace word count)
6. Top 5 memory-hungry processes:
Ans: ps aux --sort=-%mem | head -n 6 (sorts

descending, top 5)
7. Last 5 commands containing git:
Ans: history | grep git | tail -n 5 (grep filters git, tail takes 5)

Chapter 4: Interfaces
Q1. CLI vs GUI preferred situations:
• CLI: (1) Remote server SSH, (2) Batch tasks (rename 200 files in 2s), (3) Reproducible CI/CD shell scripting.
• GUI: (1) Creative visual tools (Blender, Figma), (2) Image browsing via thumbnails, (3) Wi-Fi and display setup.

Q2. Custom add command + lessons:
Script addsh: #!/bin/bash; echo "Sum: $(($1 + $2))"
Setup: chmod +x addsh ++ sudo mv addsh /usr/local/bin/add
Run: add 18 24 -> Sum: 42.
Learned: (1) Shebang picks interpreter, (2) $1, $2 read inputs, (3) chmod +x gives execute rights, (4) commands are ~~just~~ just files in $PATH.

Q3. Command options (GNU vs BSD):
• Short: -l -a or bundled -la. Long: --all, --sort (GNU double-dash).
• BSD style historically used flags without any dashes (ps aux, tar xvf).

Q4. Birth of GUI (Xerox PARC + Macintosh):

- 1970s: Xerox PARC built Xerox Alto (1973) and invented WIMP (windows, icons, menus, mouse). Xerox corporate failed to market it.
- 1979: Steve Jobs visited PARC, saw GUIs power. Apple refined it into 1984 Macintosh (overlapping windows, menu bar, $15 mouse).

Chapter 5: Operating Systems

Q1. Single CPU multitasking illusion:

- Preemptive Multitasking: OS divides CPU time into small quanta (10-20ms).
- Timer ~~interrupt~~ interrupt triggers OS -> saves Process A state into PCB -> loads Process B state from its PCB (Context Switch). Rapid switching fools human eyes.

Q2. Process memory isolation (Chrome vs VS Code):

1. Security: Web scripts in Chrome cannot inspect VS Code RAM to steal code.
2. Crash Isolation: If Chrome crashes from a bad tab, VS ~~Codde~~ Code remains alive.
3. Hardware MMU: Translates virtual to physical addresses. Unauthorized access triggers a segfault, terminating offender.

Q3. Memory allocation + ~~proccess~~ process segments:

- Allocated via Virtual Memory Paging (4KB physical frames mapped to pages).
- Layout (Low -> High): Text (code) -> Data (init globals) -> BSS (uninit globals = 0) ->

Heap (= dynamic) -> Stack (= local vars/calls).

Q4. Three college operations team analogies:
1. Memory Management <-> Room Scheduling:
OS isolates memory so programs don't overwrite each other. Ops assigns classrooms to batches. Without it: classes collide in 302.
2. CPU Scheduling <-> Shared Lab Booking:
OS time-slices CPU fairly to prevent starvation. Ops schedules AI/CPU labs. Without it: one club hogs the lab 24/7.
3. Device Drivers / HAL <-> Infrastructure Maintenance:
OS hides hardware details behind clean system calls (read/write). Ops maintains electricity, Wi-Fi, and projectors so teachers can teach.

Chapter 6. Filesystem
Q1. Environment variables + purpose:
• Named key-value pairs stored in terminal session (USER, HOME, PATH).
• ~~Purpoose~~ Purpose: Centralized runtime config without hardcoding usernames/paths.
Q2. What is PATH + command lookup:
• Colon-separated directory list (/usr/local/bin:/usr/bin:/bin).
• Shell scans folders left-to-right on entry. Executes first match.
Q3. Empty PATH, /bin/ls, + which:
• If PATH ~~si~~ is empty, bare names fail because shell has nowhere to search.
• /bin/ls still works because absolute path

bypasses $PATH lookup.
• which searches $PATH and prints the absolute path of the command.

Chapter 7: How Software Works?
Q1. Compiled vs Interpreted languages:
• Compiled (C, C++, Rust): (1) Max speed, (2) Compile-time type check, (3) Source code privacy (binary distribution).
• Interpreted (Python, JS): (1) Cross-platform, (2) Fast prototyping, (3) Line-by-line debug tracebacks.
Q2. "Translation always happens. When and who?":
• Hardware runs only binary 0s and 1s; translation is mandatory.
• Compiler: Translates Ahead of Time (AOT) to binary. Done by build tool (gcc) on developer PC. Zero runtime translation.
• Interpreter: Translates At Runtime (JIT) line-by-line during execution.

Chapter 8: Number Systems
Q1. Multiply $35_6 \times 15_6$ directly in base 6.
$3\ 5_6$
$\times\ 1\ 5_6$ [Carry: 4]

$3\ 1\ 1_6$ ($5 \times 5 = 25_{10} = 41_6$ -> 1, c 4; $5 \times 3 + 4 = 19_{10} = 31_6$)
$+\ 3\ 5\ 0_6$ ($1 \times 35_6$ shifted)

$1\ 1\ 0\ 1_6$ Ans: 1101_6 (Dec check: $23 \times 11 = 253_{10}$

$= 1101_6$?
Harder than base 10? Yes. Brain is trained for base 10 carries.
Q2. Video impact on understanding computers:
- Demystified computers: networks of physical on/off switches (transistors) forming logic gates (AND/OR/NOTs) and binary adders.
- Binary chosen for physical reliability: voltage presence vs absence.

Chapter 9: Git + Version Control

Q1. Hacktoberfest, commands, and syllabus opinion:
- Explored GitHub; used git init, clone, status, add, commit, branch, push.
- Opinion: Great in Sem 1. Avoids final_2_v2.zip habits, teaches Git early.

Q2. Git origin (Linus Torvalds, April 2005):
- Linux team used commercial BitKeeper for free until license was revoked.
- Linus wrote Git in a week in C to be distributed, fast, and branch-friendly.

Q3. "Working tree clean" meaning:
- Local folder is 100% in sync with HEAD. No modified, staged, or untracked files.
- Seen right after git commit or fresh clone. Confirms work is saved.

Chapter 10: Computer Networks

Q1. Ping command + why "no reply" != down:
- Uses ICMP Echo Request/Reply to test host reachability and latency.

• "No reply" != down: firewalls drop ICMP, but HTTP/HTTPS ports stay open.
Q2. 20B devices on 4.3B IPv4 addresses (RFC 1918 + NAT/PAT):
1. Private IP Ranges: 192.168.x.x, 10.x.x.x reused in homes worldwide.
2. NAT + PAT: Entire network shares 1 public IP via unique port numbers.
3. CGNAT: ISPs pool neighborhoods behind shared IPs, fitting 20Bt devices.